E29: the order-24 twist action on E18′s $j = 0$ line (kill test, Stage R, Stage A)

Run report and results, 2026-10-30. Note §9.27 (pre-registration, `e772a783c`) and §9.31 (results). Code and binary `0e4fccce1`. Records `experiments/34_twist24_stage_r/`. Host: the 4-vCPU container; no isolation tier; operation counts only. Under 10 core-minutes in all. **No Stage B** (no S_5 solve).

Measured

- **Kill test** (E18′s 8 curves, $p = 241 \ldots 1069$, E18′s own models): $\psi_p = (u^2 x^p, u^3 y^p)$, $u = b^{-(p-1)/6}$, is an endomorphism on all 8; its eigenvalue has order 24 ($= 4 \cdot \mathrm{ord}\ \chi_6(b)$, $\chi_6(b)$ of order 6); $|\langle \lambda_p, \lambda_\omega, -1 \rangle| = 24$; $u^2 \notin \mathbb{F}_p$ (E18′s base is not kept); a canonical model $y^2 = x^3 + b_0 t^3$ of the same order exists for every curve. Not killed.
- **Stage R** (256 walks, all correct): steps order 6 / order 24 pooled 1.701 (per curve 1.29–2.10); $\mathbb{F}_p$ cost ratio 0.743–1.202 per curve, 0.962 pooled (schoolbook), 0.884 pooled (Karatsuba): the order-24 class representative costs 191–210 per step against 109–122.
- **Stage A** (lane's `run_k4_twist24`, E18′s exact oracle; 9 arms, all logs recovered, 144 walks correct): columns cubic / twist24 = 4.00; LA ratio 13.3–15.2; r_{24} 0.0058–0.0103; modelled handover n^* $2^{135.0}$–$2^{142.2}$ (twist24) against $2^{145.7}$–$2^{151.9}$ (cubic, E18′s fold, same curves).

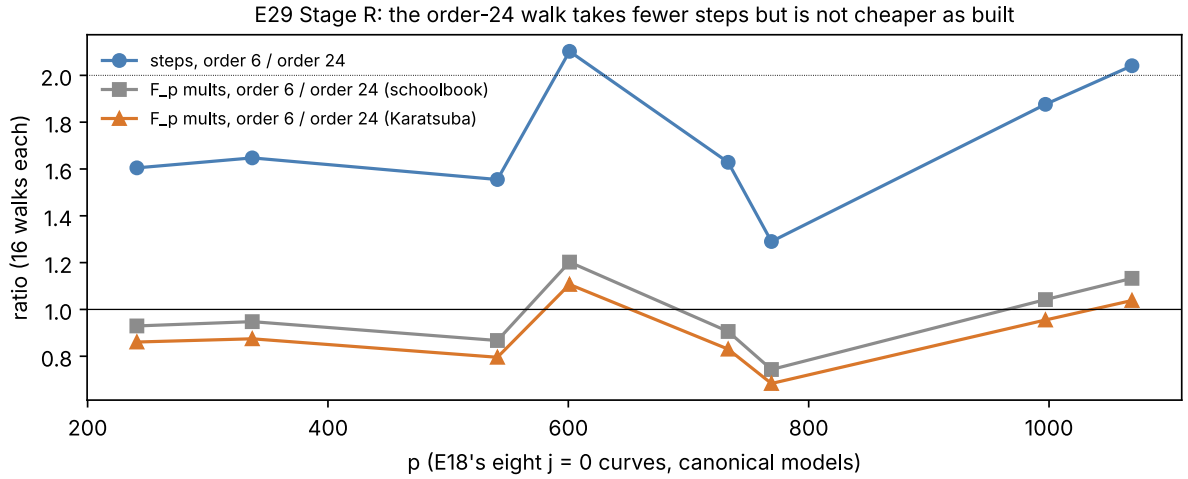


Figure 1: Stage R ratios per curve (16 walks per order). Source: `stage_r_*.json`.

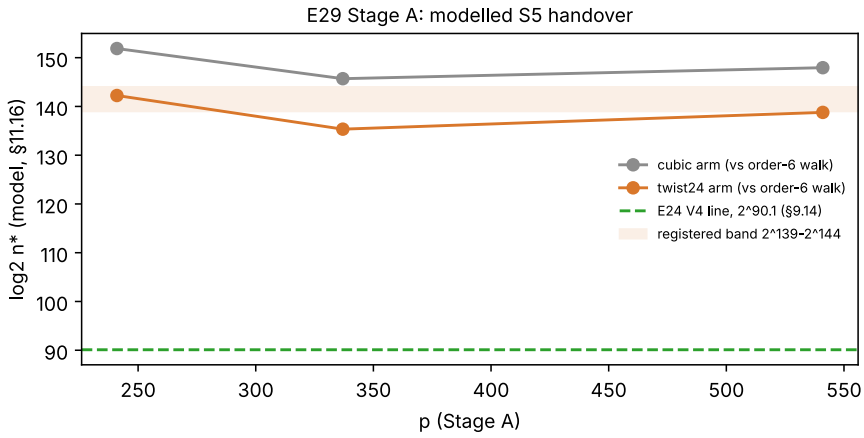


Figure 2: Stage A modelled handover (extrapolation from measured constants, $C_4 = 1.213 \cdot 10^{12}$). Source: `summary.json`.

Verdicts (§9.27)

G2.104 holds. G2.105 holds pooled (at the band's edge), falsified per curve. G2.106 falsified (order-24 walk not cheaper as built). G2.107: no correction of E18′s ψ rows at the pooled level. G2.108 holds except r_{24} (falsified low at $p = 241, 541$). G2.109 holds at $p = 241$, falsified low at 337 and 541 (and the $\geq 2^{45}$ clause misses by 0.1 bit at 337).

What it says

ψ_p has order 24 on every E18 curve. As a **walk**, the order-24 action is not a stronger reference with the lane's representative (a cheaper, abscissa-first representative is the next lever, not built). As a **base fold** it cuts E18′s $j = 0$ linear algebra ≈ 14× and moves the modelled S_5 handover ≈ 10 bits earlier: engineering on the $j = 0$ line, a model, **not an improvement of the baseline**, which sits ≥ 2^{45} earlier (E24/E25/E31). No leaderboard or `benchmarks/` row.